MANUSCRIPT

LPT: Orchestrating bounded agentic loops for automated protein binder design

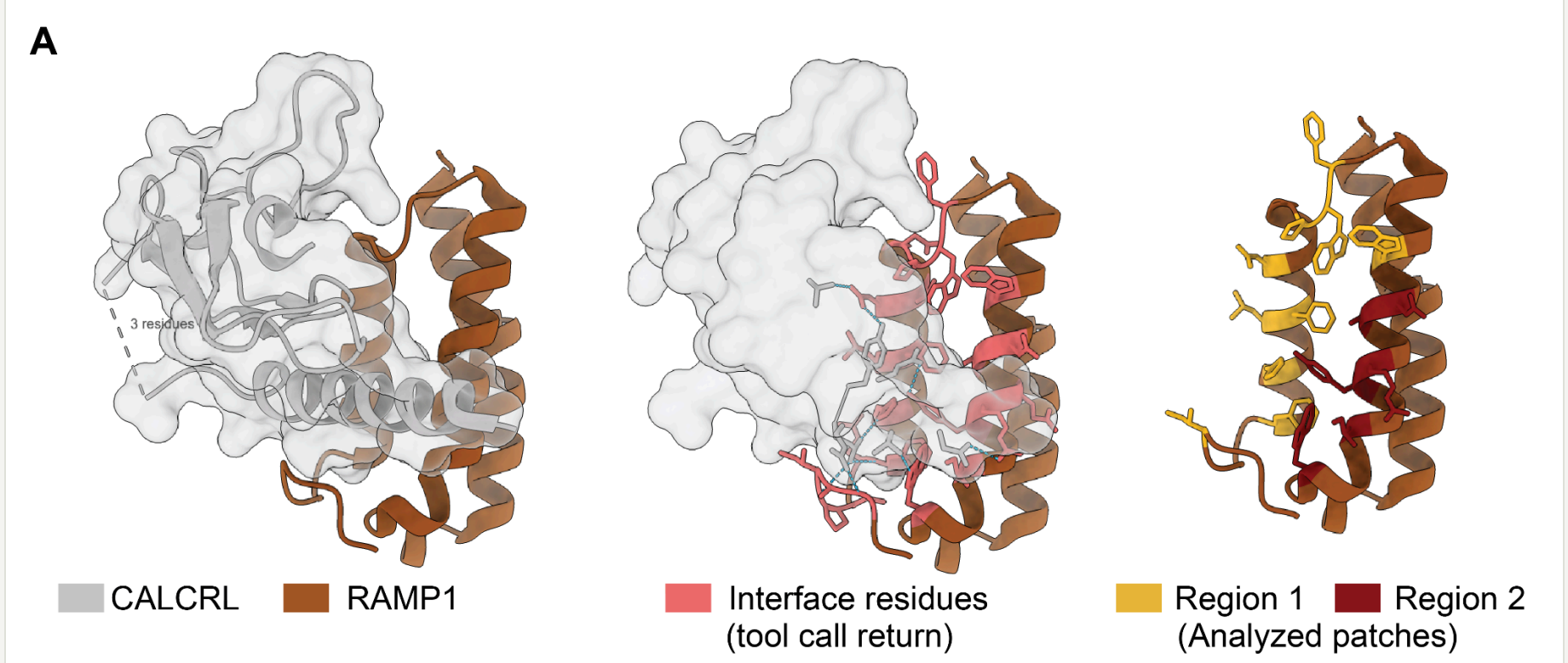
Fixed orchestration of LLM stages, each running a bounded agentic loop with its own system prompt and tool access, backed by a curated literature database. One natural-language prompt in; ranked, in silico validated binder designs out.

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Little Protein Tiger (LPT) · github.com/MichaUckelmann/little-protein-tiger

FIGURE 3A / STRUCTURE ANALYSIS

“Design novel inhibitors for pain receptors that are promising drug targets.”



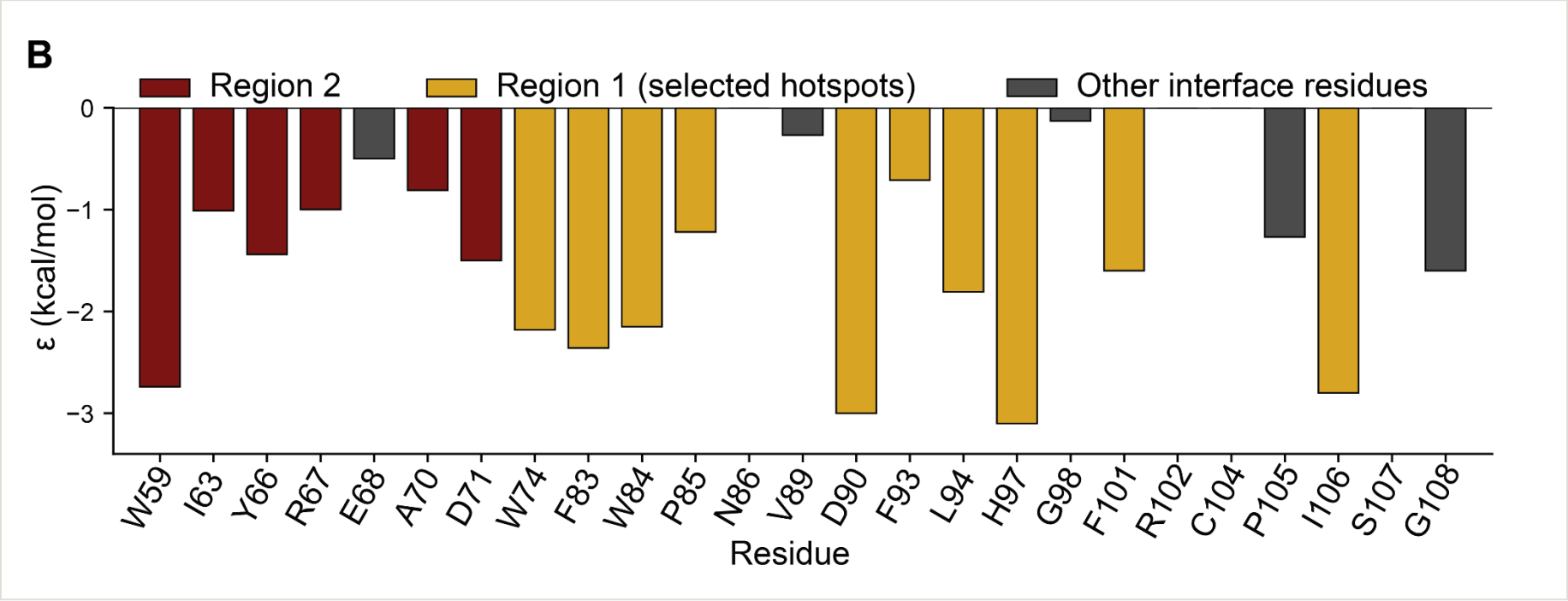
The pathway stage recommended disrupting CALCRL–RAMP1. Left: the complex selected for design (PDB 3N7S). Middle: interface residues on RAMP1 returned by the structure-analysis tool call (light red); H-bonding residues on CALCRL shown as grey sticks. Right: the two target patches analysed by the LLM stage.

PROMPT IN, STRUCTURE ANALYSIS OUT

The prompt above was the only input. Target, structure and patches were chosen by the pipeline. A code-based tool first measures the interface (buried surface area, H-bonds, per-residue contacts and an estimated binding contribution ε); the model then groups residues into clusters.

FIGURE 3B / HOTSPOT SELECTION

Per-residue binding contribution picks the hotspots.



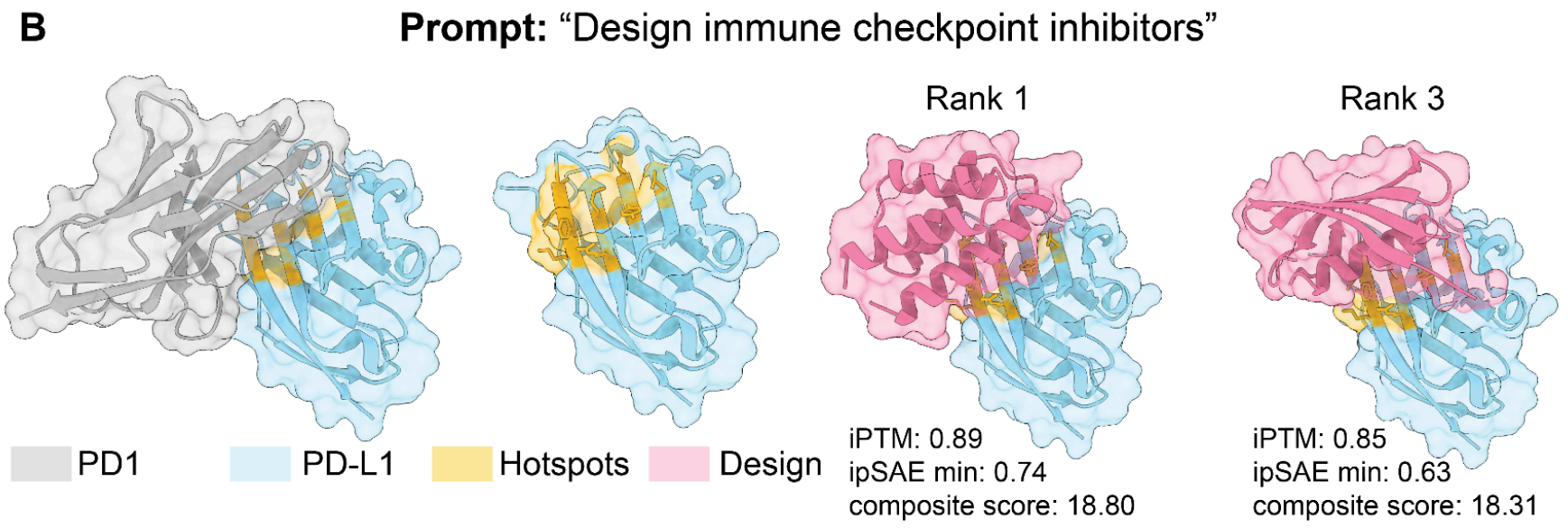
Estimated binding contribution ε (kcal/mol) returned by the structure-analysis tool for each interface residue of RAMP1. Residues of Region 1 were forwarded to the design phase as hotspots.

WHAT THE STAGE DID

The model grouped interface residues into clusters, had a second tool rate each patch (hydrophobic fraction, spatial spread, suitability), checked the choice against the literature database, and selected Region 1: a 10-residue patch with high aromatic density spanning the core heterodimerisation surface of RAMP1.

FIGURE 4B / END-TO-END RUN 2

“Design immune checkpoint inhibitors.”



The pipeline chose to disrupt PD-1/PD-L1 (PDB 4ZQK) and designed binders to PD-L1. Hotspots in gold; two highly ranked designs in pink, with ipTM, minimum ipSAE and composite score.

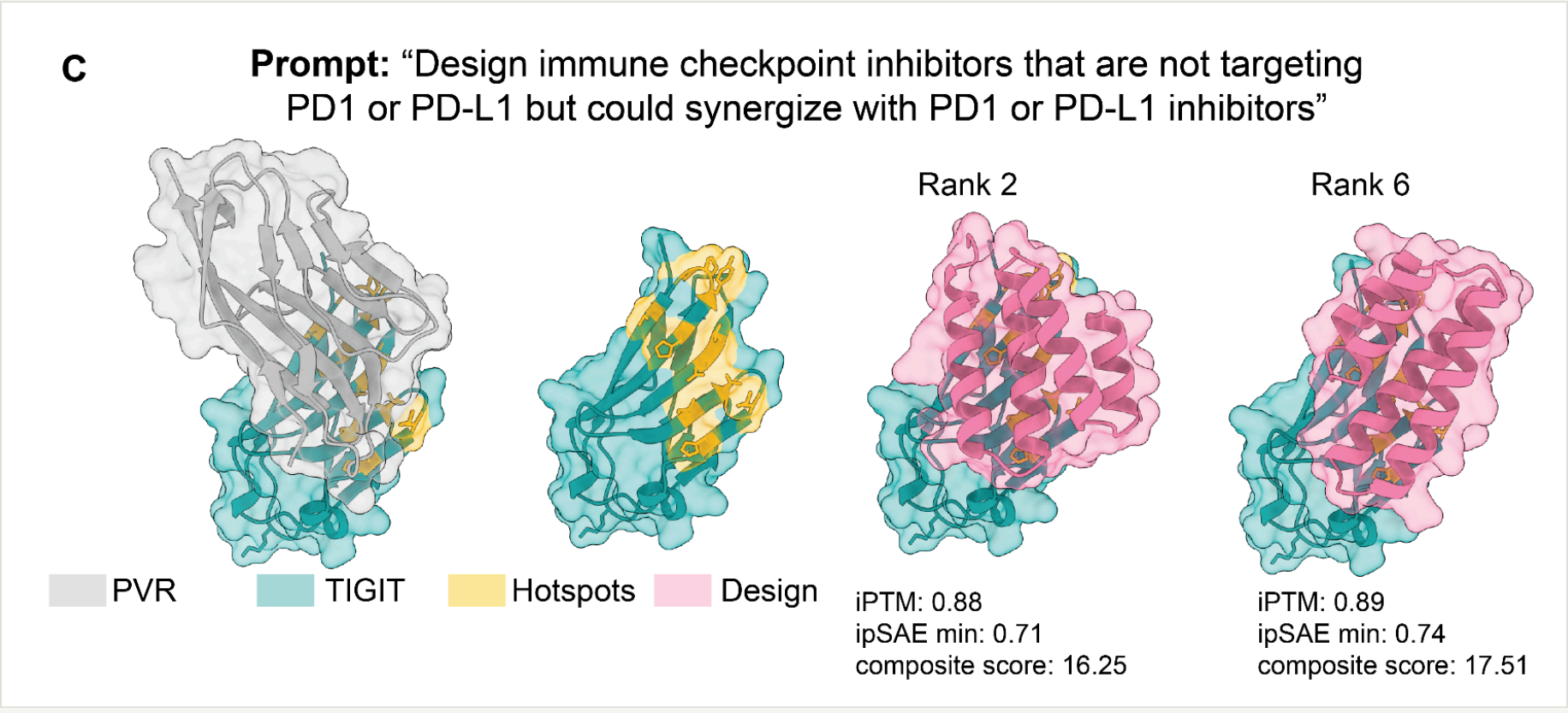
9.6%
OF REFOLDS PASS ALL GATES, IPTM ≥ 0.7

$0.42
LLM API COST

~34 h
GPU TIME, LOCAL WORKSTATION

FIGURE 4C / END-TO-END RUN 3

“...checkpoint inhibitors that are not targeting PD1 or PD-L1 but could synergize with them.”

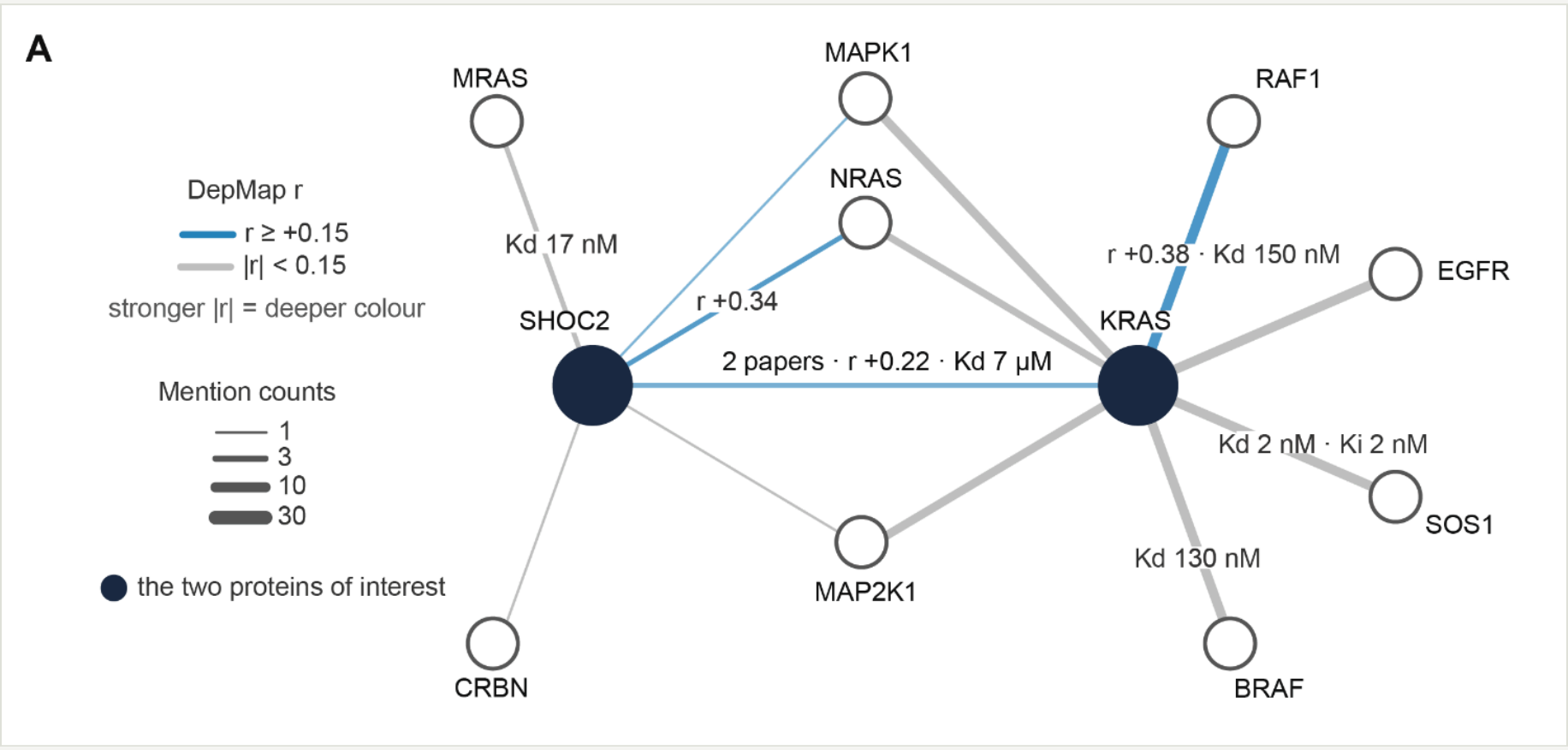


The pipeline chose to disrupt the TIGIT–PVR (CD155) interaction (PDB 3UDW). Hotspots in gold; two highly ranked designs in pink, with ipTM, minimum ipSAE and composite score.

10%	$0.39	~20 h
OF REFOLDS PASS ALL GATES, IPTM ≥ 0.7	LLM API COST	GPU TIME, LOCAL WORKSTATION

FIGURE 2A / LITERATURE DATABASE

~14,000 papers, digested and linked into an interaction graph.



Nodes are proteins; edges carry affinity information and the number of papers mentioning the interaction (edge width). Edge colour marks the DepMap CRISPR gene-effect correlation of the pair. The graph shown is a small subgraph.

CLOSED-BOOK AGAINST OPEN-BOOK EXAM

48 questions answerable from the database. With database access: 41 correct, 7 “unknown”. Without: 0 correct, 47 “unknown”, 1 wrong. The exam was biased towards database content and some questions were leading.

LITERATURE LAYER

Each paper is digested into a structured fingerprint.

Key findings with evidence, source span and confidence; proteins and structures; pathway context. Example (abridged):

USP26 functions as a negative regulator of cellular reprogramming by stabilising PRC1 complex components

DOI 10.1038/s41467-017-00301-4 · pathway_biology · experimental_in_vitro · schema 2.0

CONTEXT HOOK

In mouse embryonic fibroblasts undergoing OSKM-mediated reprogramming, the deubiquitinase USP26 physically interacts with CBX4 and CBX6, removes K48-linked polyubiquitin and protects them from proteasomal degradation. Knockdown of Usp26 increases AP+ iPSC colony generation fourfold.

KEY FINDING 1 OF 5

claim	shRNA knockdown of Usp26 enhances reprogramming efficiency fourfold
protein_pair	USP26 ↔ NANOG
evidence	~400 AP+ colonies vs ~100 in control shRNA (quantitative)
context	Dox-induced OSKM MEF reprogramming, day 12
confidence	0.95
source_span	Page 3, Para 1

KEY FINDING 3 OF 5

protein_pair	USP26 ↔ CBX6
residues	Cys304
evidence	Loss of K48 deubiquitination in C304S and Δ295–312 mutants (qualitative)
source_span	Page 5, Para 1

Fingerprints are embedded (PubMedBERT) for semantic search and aggregated into the interaction graph. Three complete examples: Supplemental Information 4.

SCOPE AND AVAILABILITY

In silico results, at low cost.

	CALCRL / RAMP1	PD-1 / PD-L1	PVR / TIGIT
Pass all gates, ipTM ≥ 0.7	17.9%	9.6%	10%
LLM API cost	$0.33	$0.42	$0.39
GPU time (local)	~11 h	~34 h	~20 h

The top 20 designs of each campaign were refolded with Boltz-2 and Protenix-v2; the majority pass all gates in all folding models.

LIMITATIONS STATED IN THE MANUSCRIPT

Experimental success rates have not been evaluated; in silico metrics are a proxy. The database (~14,000 papers in the example runs) limits target choice, and the released database is smaller because of licensing.

AVAILABILITY

Manuscript with Supplemental Information 1–6, appendices, code, skills and corpus tooling:

`github.com/MichaUckelmann/little-protein-tiger`